Lab Exercise 12 - Start and Access Kubernetes Dashboard

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Batch-2

Objective

To enable Kubernetes in Docker Desktop, deploy the Kubernetes Dashboard, and access it securely using a web browser on Windows.

Prerequisites

- Windows 10 / 11
- Docker Desktop installed
- Docker Desktop Kubernetes enabled
- Internet connection
- kubectl (comes bundled with Docker Desktop)

Step 1: Enable Kubernetes in Docker Desktop

1. Open **Docker Desktop**
2. Go to **Settings**
3. Select **Kubernetes**

4. Check **Enable Kubernetes**

5. Click **Apply & Restart**

Wait until Kubernetes status shows **Running** (green).

Step 2: Verify Kubernetes Cluster

Open **PowerShell** or **Command Prompt** and run:

- kubectl version –client

```
D:\kubs>kubectl version --client
Client Version: v1.34.1
Kustomize Version: v5.7.1
```

- Check cluster status:

- kubectl cluster-info

```
D:\kubs>kubectl cluster-info
Kubernetes control plane is running at https://127.0.0.1:56731
CoreDNS is running at https://127.0.0.1:56731/api/v1/namespaces/kube-system/services/kube-dns:dns/proxy

To further debug and diagnose cluster problems, use 'kubectl cluster-info dump'.
```

Check nodes:

kubectl get nodes

```
D:\kubs>kubectl get nodes
NAME                    STATUS   ROLES           AGE     VERSION
desktop-control-plane   Ready    control-plane   7m30s   v1.31.1
```

Expected output:

Node status should be **Ready**

Step 3: Deploy Kubernetes Dashboard

Apply the official Kubernetes Dashboard manifest:

kubectl apply -f

https://raw.githubusercontent.com/kubernetes/dashboard/v2.7.0/aio/deploy/recommended.yaml

```
D:\kubs>kubectl apply -f https://raw.githubusercontent.com/kubernetes/dashboard/v2.7.0/aio/deploy/recommended.yaml
namespace/kubernetes-dashboard created
serviceaccount/kubernetes-dashboard created
service/kubernetes-dashboard created
secret/kubernetes-dashboard-certs created
secret/kubernetes-dashboard-csrf created
secret/kubernetes-dashboard-key-holder created
configmap/kubernetes-dashboard-settings created
role.rbac.authorization.k8s.io/kubernetes-dashboard created
clusterrole.rbac.authorization.k8s.io/kubernetes-dashboard created
rolebinding.rbac.authorization.k8s.io/kubernetes-dashboard created
clusterrolebinding.rbac.authorization.k8s.io/kubernetes-dashboard created
deployment.apps/kubernetes-dashboard created
service/dashboard-metrics-scraper created
deployment.apps/dashboard-metrics-scraper created

D:\kubs>
```

Verify namespace creation:

kubectl get ns

```
D:\kubs>kubectl get ns
NAME                   STATUS   AGE
default                Active   8m14s
kube-node-lease        Active   8m14s
kube-public            Active   8m14s
kube-system            Active   8m14s
kubernetes-dashboard   Active   20s
local-path-storage     Active   8m7s
```

You should see:

kubernetes-dashboard

Step 4: Verify Dashboard Pods

Check dashboard pods:

kubectl get pods -n kubernetes-dashboard

```
D:\kubs>kubectl get pods -n kubernetes-dashboard
NAME                                         READY   STATUS    RESTARTS   AGE
dashboard-metrics-scraper-6b96ff7878-d6lnt   1/1     Running   0          55s
kubernetes-dashboard-8696f5f494-5x4pv        1/1     Running   0          55s
```

Expected status:

Running

Step 5: Create Admin User for Dashboard Access Create a

service account:

kubectl create serviceaccount dashboard-admin -n kubernetes-dashboard

```
D:\kubs>kubectl create serviceaccount dashboard-admin  -n kubernetes-dashboard
serviceaccount/dashboard-admin created
```

Create cluster role binding:

kubectl create clusterrolebinding dashboard-admin-binding --clusterrole=cluster-admin -

-serviceaccount=kubernetes-dashboard:dashboard-admin

```
D:\kubs>kubectl create clusterrolebinding dashboard-admin-binding --clusterrole=cluster-admin   --serviceaccount=kubernetes-dashboard:dashboard-admin
clusterrolebinding.rbac.authorization.k8s.io/dashboard-admin-binding created
```

Step 6: Generate Dashboard Login Token

Run the following command to get the token:

```
kubectl -n kubernetes-dashboard create token dashboard-admin
```

```
D:\kubs>kubectl -n kubernetes-dashboard create token dashboard-admin
eyJhbGciOiJSUzI1NiIsImtpZCI6IjRLckpDcHBOQ040ZW10cGg4UTJ5ajlENW1XdlhPQktIT3otbGR6REtqVm8ifQ.eyJhdWQiOlsiaHR0cHM6Ly9rdWJlcm5ldGVzLmRlZmF1bHQuc3ZjLmNsdXN0ZXIub
G9jYWwiXSwiZXhwIjoxNzcxMjY2MzMwLCJpYXQiOjE3NzEyNjI3MzAsImlzcyI6Imh0dHBzOi8va3ViZXJuZXRlcy5kZWZhdWx0LnN2Yy5jbHVzdGVyLmxvY2FsIiwianRpIjoiNmQ0ZGIzMDctZWVkNC00O
TkwLTg0MDgtMjlhY2ZhZTg2NGJhIiwia3ViZXJuZXRlcy5pbyI6eyJuYWlc3BhY2UiOiJrdWJlcm5ldGVzLWRhc2hib2FyZCIsInNlcnZpY2VhY2NvdW50Ijp7Im5hbWUiOiJkYXNoYm9hcmQtYWRtaW4iL
CJlaWQiOiJmMDc4ZTYyYy05YTA5LTQ3YjctOTU1NS0wZTUxYjE4ZjY1ZDUifX0sIm5iZiI6MTc3MTI2MjczMCwic3ViIjoic3lzdGVtOnNlcnZpY2VhY2NvdW50Omt1YmVybmV0ZXMtZGFzaGJvYXJkOmRhc
2hib2FyZC1hZG1pbiJ9.deNaoSm5QTCsR0mGgt061AiYr2F_3sqPZftS0N6WzgV8WlWXXTNy13A-fJ3kg3HHktHkHQnuOIM1F9c_-X8v70vPunx6pDirrJKiF3kMrzgqNJbr43pw7mCyw92cj-4-Efd3fDxQ
5UNtvPZfFik7CHTOnqa3894ZkW5U0fS6SCwThefxGMVv-D4XUYQLAI-cUlx18wrynzmZm_UVRYgm1Gr9Hj19ZKiHTLU7uqr9yCvATfMa9OXNnFP0zoSCVRWFT5p0U_SoBT8RCldVFqQpYMKuhnzwRqznD9xY
jCUUrUJx7IDaRcDryDVdkD0eIrXJBJSZ6bS641fMJ_UOdF2Mtw

D:\kubs>
```

Copy the generated token (you will paste it in the browser later).

Step 7: Start Kubernetes Dashboard

Run the proxy command:

```
kubectl proxy
```

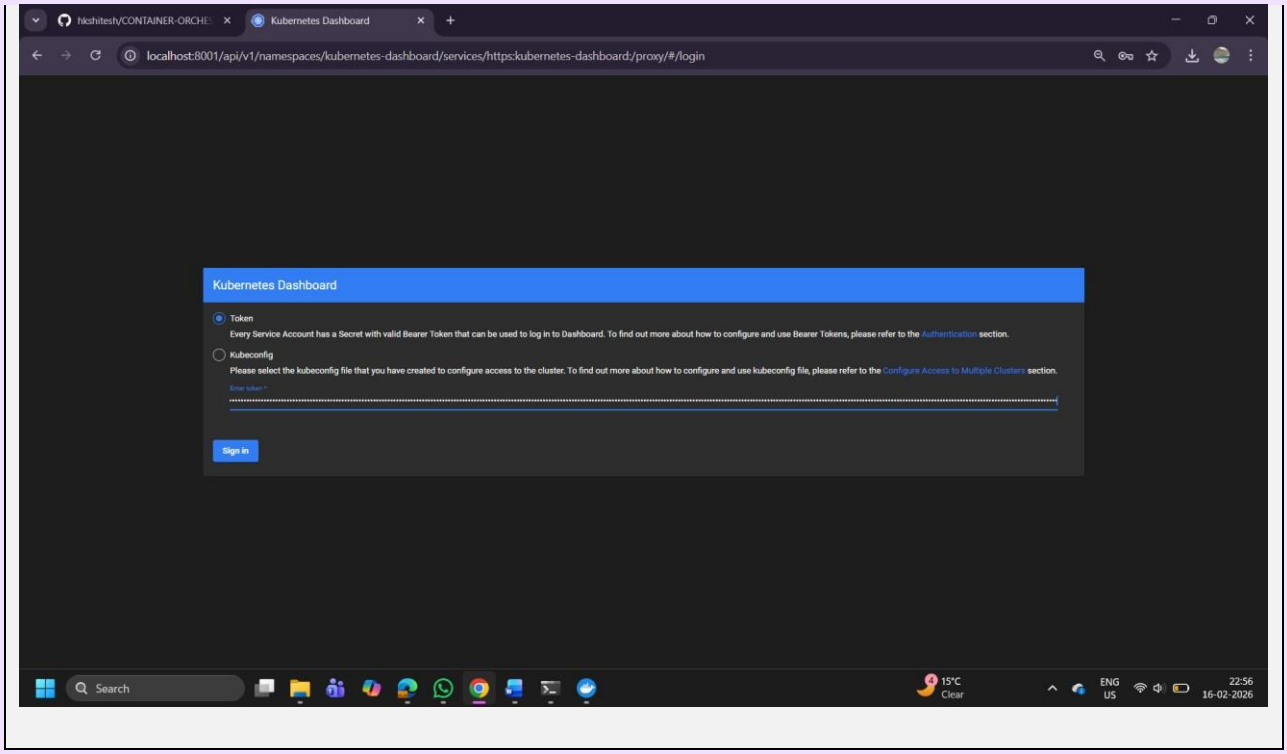
Keep this terminal **running**.

Step 8: Access Kubernetes Dashboard in Browser Open a

web browser and paste the following URL:

http://localhost:8001/api/v1/namespaces/kubernetesdashboard/services/https:kubernetes-

dashboard:/proxy/



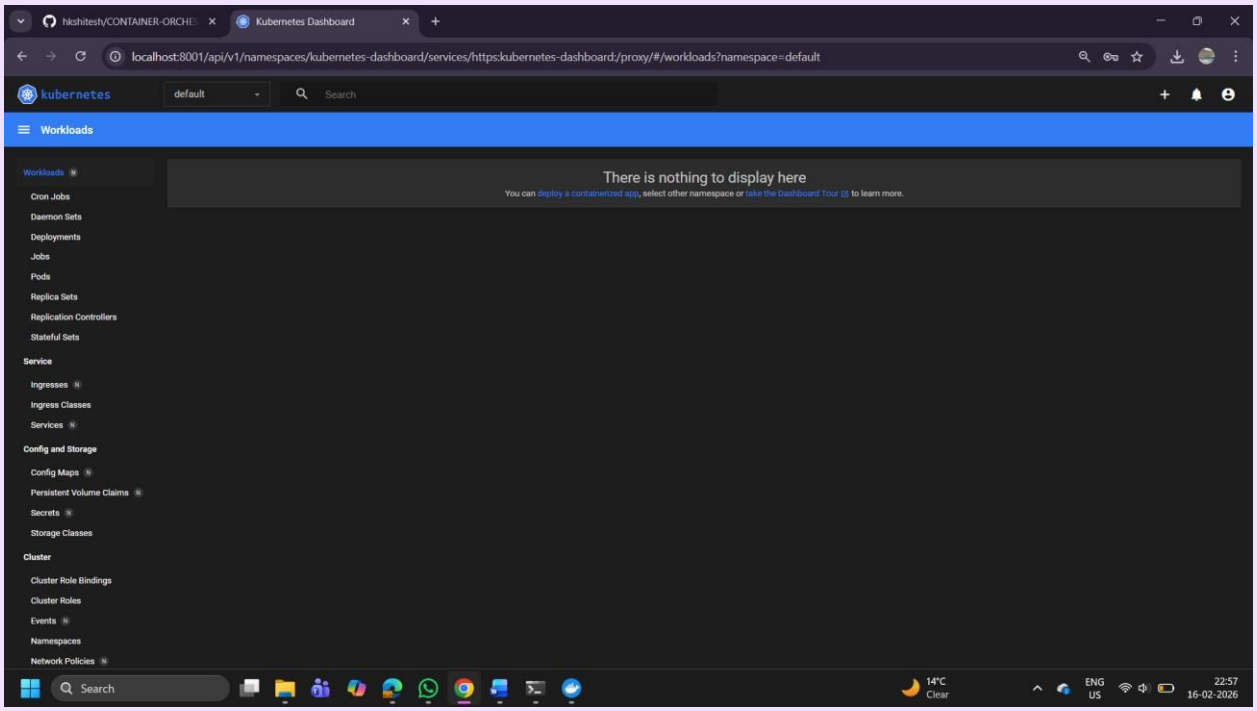
Step 9: Login to Dashboard

1. Select **Token** authentication
2. Paste the token generated earlier
3. Click **Sign In**

You should now see the **Kubernetes Dashboard UI**.

Step 10: Explore Dashboard

You can now view:

- Nodes
- Pods
- Deployments
- Services

- Namespaces
- ConfigMaps and Secrets